CMPUT 267: Machine Learning I · Fall 2026

Maximum A Posteriori Estimation

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Recall: Maximum Likelihood Estimation

Given $\mathcal{D} = (z_1, \ldots, z_n)$, choose a distribution family $p_Z(z \mid \theta)$.

MLE chooses the parameter that gives the observed data the largest likelihood:

$$\hat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta} \underbrace{\prod_{i=1}^{n} p_Z(z_i \mid \theta)}_{\mathcal{L}(\theta)}.$$

Substitute $\hat{\theta}_{\mathrm{MLE}}$ into the model to obtain the estimated distribution $p_Z(z \mid \hat{\theta}_{\mathrm{MLE}})$.

Recall: MLE for a Gaussian Mean

For $100\,\mathrm{m}^2$ houses, observe prices in **$K** ($1K = $1,000):

$$\mathcal{D} = (262, 280, 286, 300).$$

Assume independent Gaussian prices, with unknown mean and fixed $\sigma = 18$:

$$Z_i \sim \mathcal{N}(\mu^*, \sigma^2), \qquad \widehat{\mu}_{\mathrm{MLE}} = \arg\max_{\mu} \prod_{i=1}^{4} p_Z(z_i \mid \mu).$$

Gaussian MLE: Negative Log Likelihood

Substitute the Gaussian density into the likelihood:

$$\widehat{\mu}_{\mathrm{MLE}} = \arg\max_{\mu} \prod_{i=1}^{4} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(z_i-\mu)^2}{2\sigma^2}\right].$$

Take the negative log, then expand the log of each density:

$$\begin{aligned} -\log \mathcal{L}(\mu) &= -\sum_{i=1}^{4} \log\left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(z_i-\mu)^2}{2\sigma^2}\right)\right] \\ &= -\sum_{i=1}^{4} \left[-\frac{1}{2}\log(2\pi\sigma^2) - \frac{(z_i-\mu)^2}{2\sigma^2}\right] \\ &= 2\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}\sum_{i=1}^{4}(z_i-\mu)^2. \end{aligned}$$

The first term is constant in μ. Drop it when finding the minimum.

Gaussian MLE: Solving for the Mean

Recall the objective after dropping the constant:

$$\widehat{\mu}_{\mathrm{MLE}} = \arg\min_{\mu} \ \tfrac{1}{2\sigma^2} \sum_{i=1}^{4} (z_i - \mu)^2.$$

The objective is strictly convex. Set its derivative to zero:

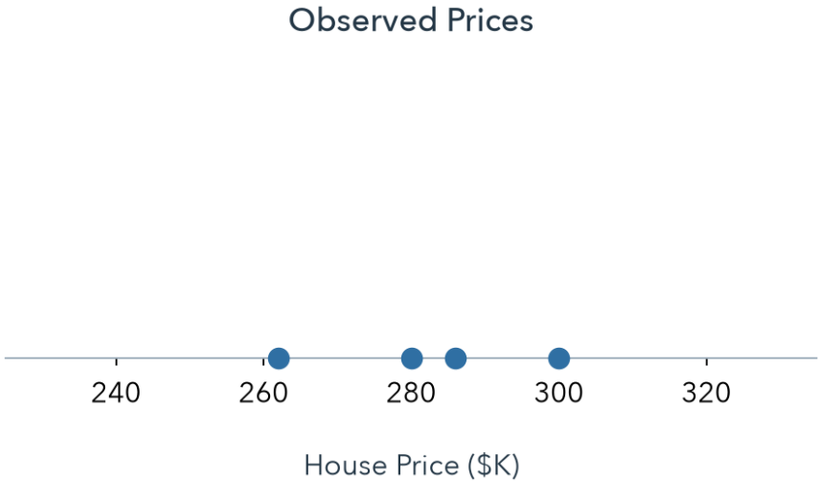
$$\begin{aligned} \tfrac{d}{d\mu}\left[\tfrac{1}{2\sigma^2} \sum_{i=1}^{4} (z_i - \mu)^2\right] &= \tfrac{1}{\sigma^2} \sum_{i=1}^{4} (\mu - z_i) \\ &= \tfrac{1}{\sigma^2}\left(4\mu - \sum_{i=1}^{4} z_i\right) = 0. \end{aligned}$$

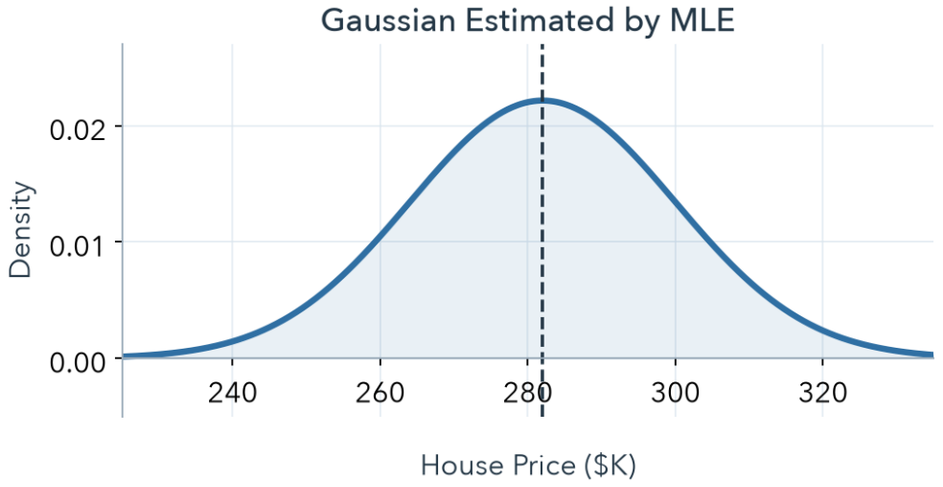
$$\widehat{\mu}_{\mathrm{MLE}} = \tfrac{1}{4} \sum_{i=1}^{4} z_i = \frac{262 + 280 + 286 + 300}{4} = 282.$$

The Distribution Estimated by MLE

Substitute the fitted mean into the Gaussian model:

$$\widehat{p}_Z(z) = p_Z(z \mid \widehat{\mu}_{\mathrm{MLE}}), \qquad \text{the density of } \mathcal{N}(282, 18^2).$$





We used the observed prices to estimate the mean, while keeping the variance fixed.

Additional Information about the Mean

MLE estimates the mean price of $100\,\mathrm{m}^2$ houses as **$282K** from our four observations.

Suppose earlier sales suggest that the mean price is around **$270K**.

We have information about the unknown parameter μ, in addition to the current data.

? How can we incorporate this information into the learning process?

This Lecture: MAP

Maximum a posteriori (MAP) estimation combines observed data with prior information about the parameters.

A Prior over the Parameter

Let μ denote the average price of houses. Suppose we know the average price μ is around $270K$, we describe it with a prior distribution $p(\mu)$.

A **prior distribution** describes our uncertainty about a parameter before observing the current data.

Individual prices: μ is a parameter controlling the center of the price distribution.

$$p(z \mid \mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(z-\mu)^2}{2\sigma^2}\right].$$

Prior over the mean: our belief about the unknown μ, centered at 270 (in \$K).

$$p(\mu) = \frac{1}{\sqrt{2\pi\tau^2}} \exp\left[-\frac{(\mu-270)^2}{2\tau^2}\right].$$

MAP: What Do We Maximize?

1. We have observed data $\mathcal{D} = (z_1, \ldots, z_n)$.
2. We represent prior knowledge about the parameter by $p(\theta)$.

The **posterior** $p(\theta \mid \mathcal{D})$ describes our uncertainty after observing the data. MAP selects its largest density value.

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} p(\theta \mid \mathcal{D}).$$

Recall MLE: What Changes with a Prior?

MAP: choose the parameter that best combines the observed data with our prior knowledge.

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} p(\theta \mid \mathcal{D}).$$

MLE: choose the parameter that best explains the observed data.

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} p(\mathcal{D} \mid \theta).$$

Bayes' Rule for Parameters

The **posterior distribution** updates our uncertainty about the parameter after observing the data. Here θ denotes a parameter such as the mean for house prices.

$$\underbrace{p(\theta \mid \mathcal{D})}_{\text{Posterior}} = \frac{\overbrace{p(\mathcal{D} \mid \theta)p(\theta)}^{\text{Likelihood Prior}}}{\underbrace{p(\mathcal{D})}_{\text{Evidence}}}.$$

- **Likelihood:** how the data depend on the candidate parameter.
- **Prior:** which parameter values we favor beforehand.
- **Evidence:** normalizes the posterior so it integrates to one.

Maximum A Posteriori Estimation

Maximum a posteriori (MAP) selects the parameter with the largest posterior density.

$$
\begin{aligned}
\hat{\theta}_{\mathrm{MAP}} &= \arg\max_{\theta} p(\theta \mid \mathcal{D}) \\
&= \arg\max_{\theta} \frac{p(\mathcal{D}|\theta)p(\theta)}{p(\mathcal{D})} \\
&= \arg\max_{\theta} p(\mathcal{D} \mid \theta)p(\theta).
\end{aligned}
$$

The evidence $p(\mathcal{D})$ is constant in θ, so it does not affect the maximizer.

Check: Likelihood or Posterior?

Two candidate parameters have the following values:

Candidate θ	Likelihood $p(\mathcal{D} \mid \theta)$	Prior density $p(\theta)$	Product $p(\mathcal{D} \mid \theta)p(\theta)$
θ_A	0.08	0.2	0.016
θ_B	0.05	0.6	0.030

? Among these two candidates, which does MLE select? Which does MAP select?

MLE selects θ_A; MAP selects θ_B.

Trick: Apply the Negative Log

Given θ, assume the observations in $\mathcal{D} = (z_1, \ldots, z_n)$ are i.i.d.

$$
\begin{aligned}
\hat{\theta}_{\text{MAP}} &= \arg\max_{\theta} p(\mathcal{D} \mid \theta) p(\theta) \\
&= \arg\max_{\theta} \left[\prod_{i=1}^{n} p_Z(z_i \mid \theta) \right] p(\theta) \\
&= \arg\min_{\theta} \left\{ -\log \left[\left(\prod_{i=1}^{n} p_Z(z_i \mid \theta) \right) p(\theta) \right] \right\} \\
&= \arg\min_{\theta} \left[-\sum_{i=1}^{n} \log p_Z(z_i \mid \theta) - \log p(\theta) \right].
\end{aligned}
$$

Minimize **negative log likelihood + negative log prior**. There are n data terms and **one prior term**.

MAP for a Gaussian Mean

MAP for Gaussian Mean: House Prices

Return to our houses price example. We now derive the MAP estimate of the mean price μ (in \$K).

$$Z_i \mid \mu \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2), \qquad \mu \sim \mathcal{N}(270, \tau^2).$$

- $Z_i \mid \mu \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$: the hourse prices follow a gaussian distribution of mean μ, variance σ^2.
- $\mu \sim \mathcal{N}(270, \tau^2)$: the mean of the house prices distribution follow another gaussian distribution.

MAP for Gaussian Mean: Negative Logs

Recall the observation density and the prior density:

$$p_Z(z_i \mid \mu) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(z_i-\mu)^2}{2\sigma^2}}, \qquad p(\mu) = \frac{1}{\sqrt{2\pi\tau^2}} e^{-\frac{(\mu-270)^2}{2\tau^2}}.$$

Recall the MAP minimization form, then substitute these densities:

$$\begin{aligned}
\widehat{\mu}_{\text{MAP}} &= \arg\min_{\mu} \left[-\sum_{i=1}^{n} \log p_Z(z_i \mid \mu) - \log p(\mu) \right] \\
&= \arg\min_{\mu} \left[\tfrac{n}{2}\log(2\pi\sigma^2) + \tfrac{1}{2\sigma^2} \sum_{i=1}^{n} (z_i - \mu)^2 + \tfrac{1}{2}\log(2\pi\tau^2) + \tfrac{(\mu-270)^2}{2\tau^2} \right] \\
&= \arg\min_{\mu} \left[\cancel{\tfrac{n}{2}\log(2\pi\sigma^2)} + \tfrac{1}{2\sigma^2} \sum_{i=1}^{n} (z_i - \mu)^2 + \cancel{\tfrac{1}{2}\log(2\pi\tau^2)} + \tfrac{(\mu-270)^2}{2\tau^2} \right].
\end{aligned}$$

MAP for Gaussian Mean: Objective

Add the negative log likelihood and negative log prior, then drop constants in μ.

$$\widehat{\mu}_{\mathrm{MAP}} = \arg\min_{\mu} \left[\underbrace{\frac{1}{2\sigma^2} \sum_{i=1}^{n} (z_i - \mu)^2}_{\text{Data term}} + \underbrace{\frac{(\mu - 270)^2}{2\tau^2}}_{\text{Prior term}} \right].$$

The data favor the sample mean $\bar{z}$. The prior favors 270.

MAP for Gaussian Mean: Solution

Recall the objective:

$$\widehat{\mu}_{\mathrm{MAP}} = \arg\min_{\mu} \left[\frac{1}{2\sigma^2} \sum_{i=1}^{n} (z_i - \mu)^2 + \frac{(\mu - 270)^2}{2\tau^2} \right].$$

This objective is strictly convex. Differentiate and set the result to zero:

$$\frac{d}{d\mu} \left[\frac{1}{2\sigma^2} \sum_{i} (z_i - \mu)^2 + \frac{(\mu - 270)^2}{2\tau^2} \right] = \frac{n\mu - \sum_i z_i}{\sigma^2} + \frac{\mu - 270}{\tau^2} = 0.$$

$$\left(\frac{n}{\sigma^2} + \frac{1}{\tau^2} \right) \mu = \frac{n\bar{z}}{\sigma^2} + \frac{270}{\tau^2}.$$

Understanding MAP: Weighted Average

MAP is a weighted average of the data mean and prior mean.

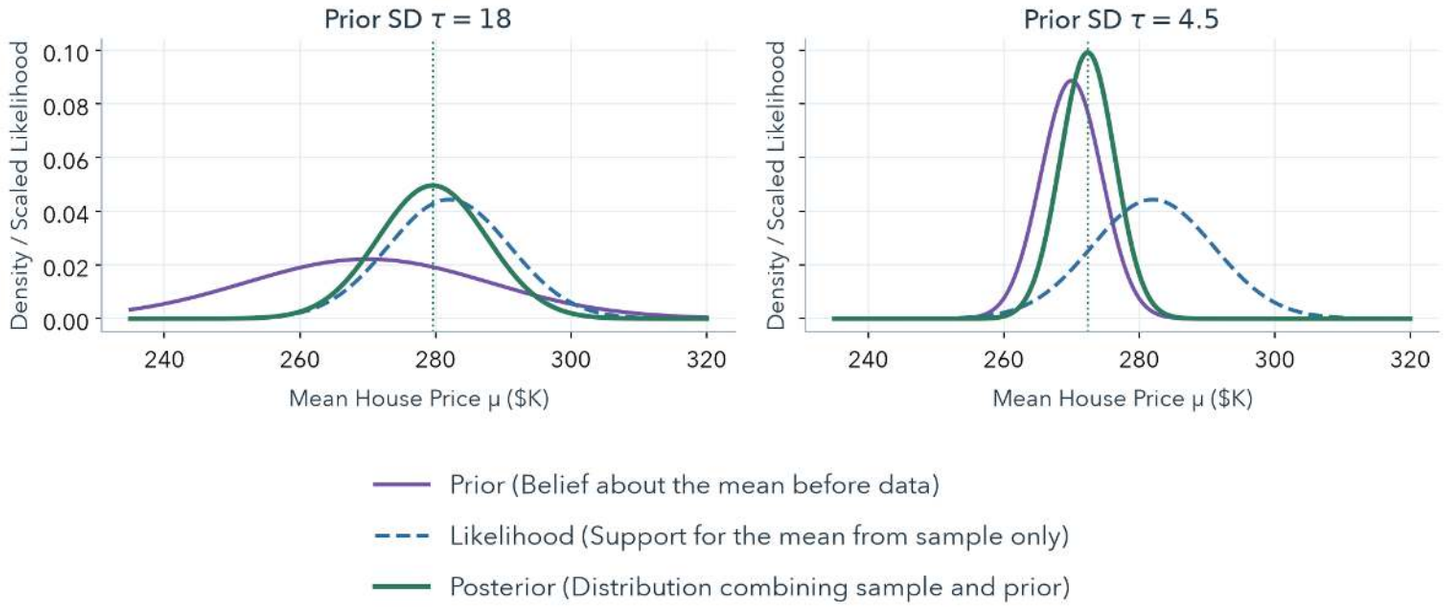
$$\widehat{\mu}_{\text{MAP}} = \frac{(n/\sigma^2)\bar{z} + (1/\tau^2)\,270}{n/\sigma^2 + 1/\tau^2}.$$

- $\bar{z} = \frac{1}{n}\sum_{i=1}^{n} z_i$: the **data mean**, with weight n/σ^2.
- 270: the **prior mean**, with weight $1/\tau^2$.

- More data n increases the weight on the data mean.
- Smaller prior variance τ^2 (stronger prior belief) increases the weight on the prior mean.

Prior, Likelihood, and Posterior

Same four house prices and prior mean of $270K; different prior uncertainty.



A narrower prior pulls the posterior mode closer to $270K.

MAP Learner

MAP Learner: The Task

Given observed feature-label pairs

$$\mathcal{D} = \big((\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\big),$$

For each input $\mathbf{X} = \mathbf{x}_i$, assume a Gaussian conditional distribution:

$$Y \mid \mathbf{X} = \mathbf{x}_i, \mathbf{w} \sim \mathcal{N}(\mathbf{w}^\top \mathbf{x}_i, \sigma^2).$$

This is Gaussian with mean $\mathbf{x}_i^\top \mathbf{w}$ and fixed variance $\sigma^2 > 0$.

Additionally, specify a prior $p(\mathbf{w})$ over the weights.

A Prior on the Regression Weights

Suppose we prefer smaller non-intercept weights, as in regularization.

Assign independent Gaussian priors to the non-intercept weights:

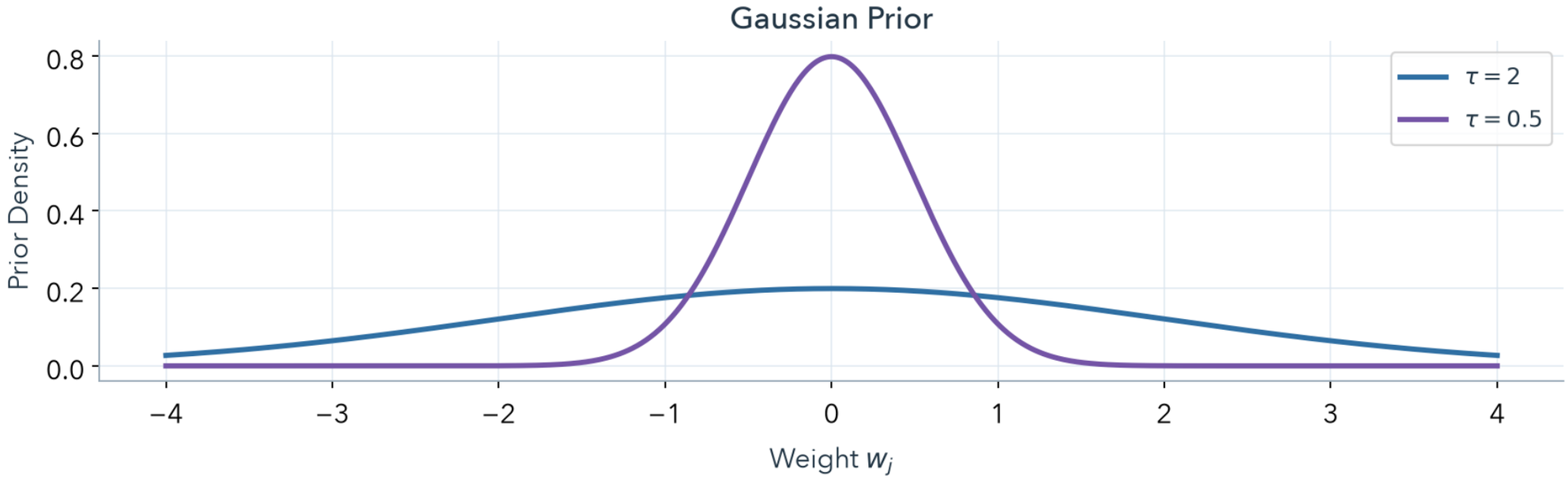
$$w_j \sim \mathcal{N}(0, \tau^2), \qquad j = 1, \ldots, d.$$

τ^2 describes prior uncertainty about each weight. Smaller τ^2 puts more density near zero.

MAP needs two distributions:

- $p(y \mid \mathbf{x}, \mathbf{w})$: the conditional distribution of **labels**, given input and weights.
- $p(\mathbf{w})$: the prior distribution over **weights**.

Gaussian Priors and Weight Size



A smaller prior SD puts more density near zero: we expect smaller weights.

MAP Learner: The Objective

Recall: maximize conditional likelihood times the weight prior.

$$
\begin{aligned}
\widehat{\mathbf{w}}_{\text{MAP}} &= \arg\max_{\mathbf{w}} \left[\prod_{i=1}^{n} p_{Y|\mathbf{X}}(y_i \mid \mathbf{x}_i, \mathbf{w})\right] p(\mathbf{w}) \\
&= \arg\min_{\mathbf{w}} \left\{-\log\left[\left(\prod_{i=1}^{n} p_{Y|\mathbf{X}}(y_i \mid \mathbf{x}_i, \mathbf{w})\right) p(\mathbf{w})\right]\right\} \\
&= \arg\min_{\mathbf{w}} \left[-\sum_{i=1}^{n} \log p_{Y|\mathbf{X}}(y_i \mid \mathbf{x}_i, \mathbf{w}) - \log p(\mathbf{w})\right].
\end{aligned}
$$

The objective is **negative log likelihood + negative log prior**.

Gaussian MAP Gives Ridge Regression

For the Gaussian label model and independent Gaussian weight priors:

$$-\sum_{i=1}^{n} \log p(y_i \mid \mathbf{x}_i, \mathbf{w}) = \underbrace{\tfrac{n}{2}\log(2\pi\sigma^2)}_{\text{Constant}} + \tfrac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mathbf{x}_i^\top \mathbf{w})^2,$$

$$-\log p(\mathbf{w}) = \underbrace{\tfrac{d}{2}\log(2\pi\tau^2)}_{\text{Constant}} + \tfrac{1}{2\tau^2}\sum_{j=1}^{d} w_j^2.$$

Drop constants in $\mathbf{w}$ and combine the two terms:

$$\widehat{\mathbf{w}}_{\text{MAP}} = \arg\min_{\mathbf{w}} \left[\tfrac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - \mathbf{x}_i^\top \mathbf{w})^2 + \tfrac{1}{2\tau^2} \sum_{j=1}^{d} w_j^2 \right].$$

MAP Learner: The Ridge Objective

Multiply the **whole objective** by $2\sigma^2/n$:

$$\widehat{\mathbf{w}}_{\text{MAP}} = \arg\min_{\mathbf{w}} \left[\frac{1}{n} \sum_{i=1}^{n} (y_i - \mathbf{x}_i^\top \mathbf{w})^2 + \frac{\sigma^2}{n\tau^2} \sum_{j=1}^{d} w_j^2 \right].$$

Set $\lambda = \sigma^2/\tau^2$. This is $\widehat{L}_D(\mathbf{w}) + \frac{\lambda}{n} \sum_{j=1}^{d} w_j^2$, the **ridge objective**.

Interpreting Regularization Strength

σ^2: variance of the conditional label distribution $p(y \mid \mathbf{x}, \mathbf{w})$.

τ^2: variance of each Gaussian weight prior $p(w_j)$.

With fixed observation variance, $\lambda = \sigma^2 / \tau^2$.

Prior assumption	Regularization	Effect
Large τ^2	Small λ	Weak pull toward zero
Small τ^2	Large λ	Strong pull toward zero

Other Priors and Regularization

A Different Prior: Laplace

A **Laplace distribution** is symmetric around its center, with an exponential decrease on either side.

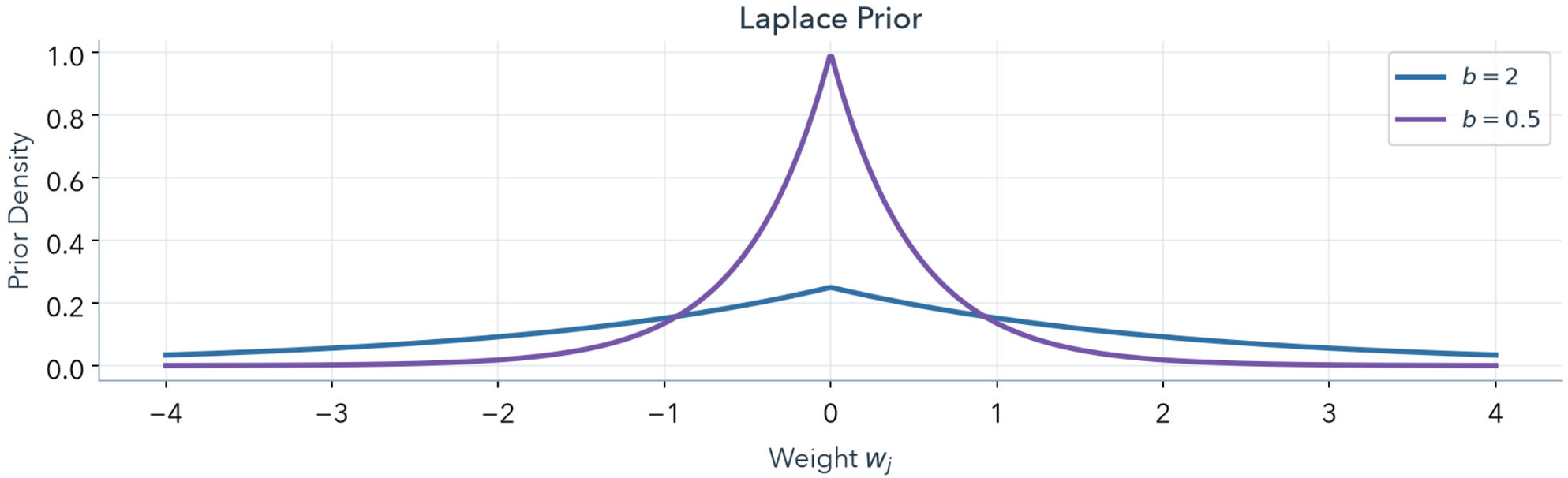
For a zero-centered weight prior with scale $b > 0$:

$$p(w_j) = \frac{1}{2b} \exp\left(-\frac{|w_j|}{b}\right).$$

Assume independent priors for $w_1, \ldots, w_d$; leave the intercept unpenalized.

b controls the spread of the prior; its variance is $2b^2$.

Laplace Prior and Weight Size



A smaller scale b puts more density near zero: we expect smaller weights.

Lasso: The MAP Objective

Recall: maximize conditional likelihood times the weight prior.

$$
\begin{aligned}
\widehat{\mathbf{w}}_{\text{MAP}} &= \arg\max_{\mathbf{w}} \left[\prod_{i=1}^{n} p_{Y|\mathbf{X}}(y_i \mid \mathbf{x}_i, \mathbf{w})\right] p(\mathbf{w}) \\
&= \arg\min_{\mathbf{w}} \left\{-\log\left[\left(\prod_{i=1}^{n} p_{Y|\mathbf{X}}(y_i \mid \mathbf{x}_i, \mathbf{w})\right) p(\mathbf{w})\right]\right\} \\
&= \arg\min_{\mathbf{w}} \left[-\sum_{i=1}^{n} \log p_{Y|\mathbf{X}}(y_i \mid \mathbf{x}_i, \mathbf{w}) - \log p(\mathbf{w})\right].
\end{aligned}
$$

The objective is **negative log likelihood + negative log prior**.

Laplace MAP: Negative Logs

Keep the Gaussian label model; replace the weight prior with Laplace:

$$-\sum_{i=1}^{n} \log p(y_i \mid \mathbf{x}_i, \mathbf{w}) = \underbrace{\tfrac{n}{2}\log(2\pi\sigma^2)}_{\text{Constant}} + \tfrac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mathbf{x}_i^\top \mathbf{w})^2,$$

$$-\log p(\mathbf{w}) = \underbrace{d\log(2b)}_{\text{Constant}} + \tfrac{1}{b}\sum_{j=1}^{d}|w_j|.$$

Drop constants in $\mathbf{w}$ and combine the two terms:

$$\widehat{\mathbf{w}}_{\text{MAP}} = \arg\min_{\mathbf{w}} \left[\tfrac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mathbf{x}_i^\top \mathbf{w})^2 + \tfrac{1}{b}\sum_{j=1}^{d}|w_j| \right].$$

MAP Learner: The Lasso Objective

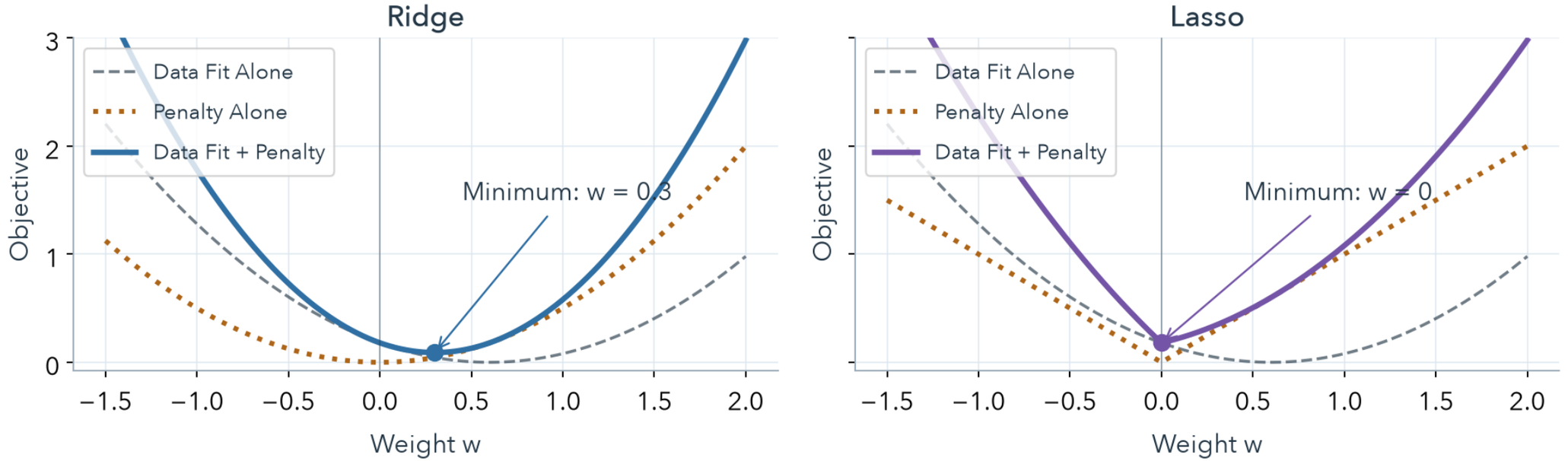
Multiply the **whole objective** by $2\sigma^2/n$:

$$\widehat{\mathbf{w}}_{\text{MAP}} = \arg\min_{\mathbf{w}} \left[\frac{1}{n} \sum_{i=1}^{n} (y_i - \mathbf{x}_i^\top \mathbf{w})^2 + \frac{2\sigma^2}{nb} \sum_{j=1}^{d} |w_j| \right].$$

Set $\lambda_1 = 2\sigma^2/b$. This is $\widehat{L}_D(\mathbf{w}) + \frac{\lambda_1}{n} \sum_{j=1}^{d} |w_j|$, the **lasso objective**.

Ridge and Lasso: Why Zero?

One-weight example: the data alone prefer $w = 0.6$.



Ridge: a smooth minimum near zero. **Lasso:** a corner that can hold the minimum at zero.

Wrap-up: MLE, MAP, and Regularization

Estimation method	Observation model	Parameter prior	Result
MLE	Gaussian	–	Least squares
MAP	Gaussian	Gaussian slopes	Ridge
MAP	Gaussian	Laplace slopes	Lasso

Negative log likelihood determines the data-fitting term. **Negative log prior** determines the penalty.